Electric Power Systems Research 230 (2024) 110223

Contents lists available at ScienceDirect

Electric Power Systems Research

journal homepage: www.elsevier.com/locate/epsr

A detailed review of power system resilience enhancement pillars

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ARTICLE INFO

Keywords:
Building resilience
Distributing
Enhancement pillars
Hardening, Smartening

ABSTRACT

The increasing occurrence of extreme vulnerabilities to electric grids due to natural catastrophes and cyber-attacks, has resulted in a rise in global power outages. This has compelled a sense of urgency to investigate resiliency issues and implement risk mitigation strategies. This article conducts a state-of-the-art review of the key pillars of power system resilience, examining the concepts and techniques to improve resilience against low-probability, high-impact events. The highlighted pillars include smartening, hardening, distributing, building, and the corresponding enhancement techniques. The article begins by enlightening the origin of resilience and its relevance in the power system context, followed by an exploration of the four key pillars. Additionally, it investigates the quantification of resilience in different levels of power systems facing diverse sources of extreme events. Emerging technologies that played a crucial role in enhancing resilience are discussed. Furthermore, the paper explores the concept of Greenfield and Brownfield power system design approaches to improve resilience against extreme events. Moreover, a transformative idea extending the conventional resilience philosophy, called antifragility, is clearly stated. The discussion extends to the need for resilience study from global perspectives, state-of-the-art technologies, and challenges and opportunities. To conclude, this review sheds light on the emerging trend of resilience research, paving the way for future research directions.

1. Introduction

In the 21st century, the energy transition is crucial in maintaining reliable, resilient, and secure modern power system operations [1]. This requires building much more modern power systems infrastructures [2]. After extreme power failure events, the power system recovery cost is exceptionally high, which directly impacts the global economy. On the other hand, the N-1 design criteria for power systems are not always defended against extreme events due to the increasing frequency of natural disasters and cyber-physical attacks [3]. Taking into consideration recovery costs, economy, and system robustness, it has been a challenging effort to build a resilient electric infrastructure [4], which helps in maintaining the uninterruptible power supply during normal operating conditions, quick load restoration after the events, increased vulnerability to disaster and cyber-attack [5], damage prevention, survivability, etc. [6]. Further, to address these challenges, several foresight of power outages due to extreme events in different circumstances are to be realized [7]. In the last few years, most research on power outages has highlighted the study of reliability. To understand the concept of reliability, two critical aspects are to be measured: adequacy and security [8]. Adequacy represents the uninterruptible electricity supply to the consumers, whereas security represents the ability to cope with power system faults [9]. However, the reliable design-based power system doesn't consider extreme events like natural disasters and cyber-attacks. To consider extreme events, the power system should be reconfigured to make it more robust, which is called a resilient system.

Notably, the electric infrastructure needs to be resilient to cope with unforeseen events, so the concept of resilience is vital. The origin of resilience comes from the Latin "*resilio*", which states 'spring back' [10]. But, according to the dictionary meaning, this suggests the ability to restore quickly from extreme events. While a diversity of definitions of the term 'resilience' are suggested, in 1973, Holing et al. defined the term resilience, i.e., a measure of the persistence of systems and their ability to absorb change and disturbance while maintaining *the same relationships between populations or state variables*. This still upholds the undistinguishable communications between populations and state variables [11]. From the perspective of the power system, resilience can be

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https://doi.org/10.1016/j.epsr.2024.110223
Received 4 September 2023; Received in revised form 17 January 2024; Accepted 2 February 2024
Available online 12 February 2024
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Nomenclature

Abbreviation

ACE	Area control error
AGC	Automatic generation control
AM	Advancing metering
CVaR	Conditional value-at-risk
DC	Direct current
DOE	Department of Energy
DSM	Demand side management
DS	Distribution System
EES	Electrical energy storage
EENS	Expected energy not served
EPRI	Electric Power Research Institute
EV	Electric vehicles
FACTs	Flexible alternating current transmission systems
HILP	High probability, low impact
ICT	Information and communication technology
IMG	Isolated microgrid
IoT	Internet of things
LOLF	Loss of load frequency
LOLE	Loss of load expectation
LOLP	Loss of load probability
MILP	Mixed integer linear programming
MCS	Monte carlo simulation
MG	Microgrid
MMG	Multi-Microgrid
N	Total number of critical components used in power systems
−1	The system must be able to cope with the failure of one of these critical components.
OPF	Optimal power flow
PowDDER	Power distribution designing for resilience application
PMU	Phase measurement unit
RTU	Remote terminal unit
R&D	Research and development
SAPHIRE	Systems analysis program for hands-on integrated reliability evaluations
SSB	Solid-state battery
SST	Solid-state transformer
SG	Smart grid
SCADA	Supervisory control and data acquisition
VPP	Virtual power plant
VOLL	Value of loss of load
4R	Robustness, Redundancy, Rapidness, and Resourcefulness
WRAP	Withstand, Respond, Adapt, and Prevent

Definitions

Antifragility	Beyond resilience: Refers to attributes of systems that benefit from events.
Brownfield	Refers to the approach to redevelopment of projects in existing land.
Energy trilemma index	Refers to three factors: Energy sustainability, security, and equity
Greenfield	Refers to an approach with a new project in an underdeveloped land.
PowDDER	Adaptive capacity measurement software
Reliability	Refers to the ability of the system to maintain the continuity of power supply.
Resiliency	Refers to the capability to immediately recover from disruptive events.
Robustness	Refers to withstanding capability that can cope with extreme events
Rapidness	Faster approach, contingency plans, and emergency operation
Redundancy	Refers to the ability to manage the power supply using a backup component or system.
FLEP	Refers to resilience index-Fast, low, extensive, and promptly behave
SAPHIRE	Probabilistic risk assessment tool
Stability	Refers to the ability of a system to maintain a balanced operation.

defined as the ability to withstand and restore normal operation with minimal time to minimize the disaster effect during and after unforeseen events [2]. However, research in the power system has emphasized the terms reliability and resiliency, and researchers are moving forward to explore the critical concerns of reliability and resiliency [12]. A few important factors, such as robustness, cost-effectiveness, reliability, and security, can measure the power system's resiliency. Although the terms reliability and resiliency are often viewed as the same, these are different according to the probability of occurrence and impact, as shown in Fig. 1. Reliability is concerned with normal operating conditions or controllable fault conditions. In contrast, resiliency is linked with high-impact and low-probability (HILP) events, like natural disasters, cyber-attacks, and uncontrollable faults. Other than the resilience triangle and trapezoid, a few other studies have used the slope of the performance curve to measure the system's resilience, such as conceptual resilience curve [13], a measure of performance [14], quality performance [15], performance degradation period [16].

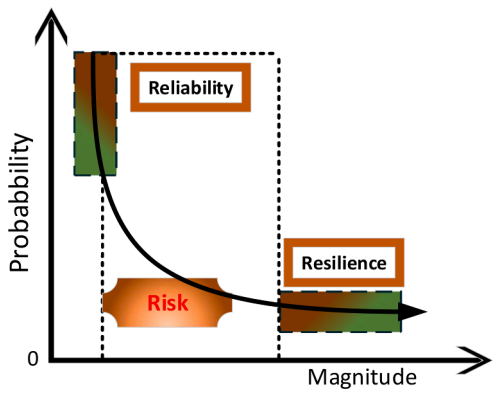


Fig. 1. Reliability vs. Resiliency.

There are various ways to represent the resilience curve that signifies the ways of functioning in different states, for example, resilience triangle [17], resilience trapezoid [18], and multi-phase resilience trapezoid [19]. Many researchers have reported the resilience triangle, where two distinct states have been presented, as shown in Fig. 2. It considers two main factors, *i.e.*, robustness and rapidness, where robustness is defined as the withstanding ability to cope against extreme events. Rapidness refers to the time from the points of power failure to the final restoration, which needs to be faster to avoid power outages. Eventually, the resilience achievement can be appraised as robustness over rapidness [17]. Nevertheless, abundant research has been carried out to understand the resilience concept, such as withstanding any extreme events, responding quickly for restoration, making a more adaptable system through resourcefulness, and preventing future occurrences, as presented in Fig. 3, and its characteristics are highlighted in Fig. 4. Moreover, withstanding refers to the robustness of the system, which signifies the coping capacity in response to extreme events. Further, the respond refers to the rapidness of the system, by which the quick

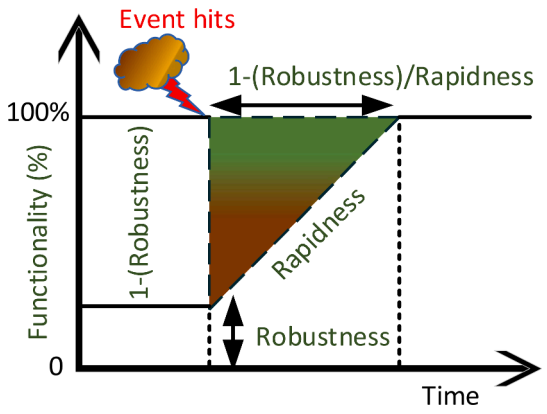


Fig. 2. Resilience triangle [17].

restoration model can be developed to minimize the time and cost of recovery. In the adapt phase, a controllable and adaptable system can be realized by controlling/restoring the operation automatically after any event. Finally, future occurrences can be minimized in the 'prevent' phase by measuring the probability of failures and comparing them with the past event data. With this, the system's resiliency can be outlined as enhancing the infrastructural strength to resist hazardous events, simultaneously responding to any damage, and adapting to tackle any future untoward events, as depicted in Fig. 4. Moreover, resilience can also be defined into four factors: robustness, redundancy, rapidness, and resourcefulness, and their characteristics are seen in TABLE I. Together, these four factors build an interactive framework that improves resiliency, enabling it to navigate disturbances and sustain reliable operation in the face of HILP events.

Over the past few decades, natural disasters are increasing worldwide, which are hurricanes, earthquakes, storms, floods, cyber-attacks, equipment failures, and other power system faults, as shown in Figs. 5 and 6. Fig. 5 presents the number of people affected by a major power outage due to the extreme events from 1999 to 2019 [20]. Further, Fig. 6 clearly shows an understanding of various events in the last twenty years. Most power outages happen because of equipment failure and natural disasters. As the power system is the keystone of smart cities in modern societies and the main pillar of the global economy and social life, it should be taken care of in the form of robust design and resilience consideration to deal with extreme events. For that, the resilient power system is of extreme importance nowadays. A resilient power system means the continuity of power supply, even after/during natural disasters [21]. To have a complete understanding of power system resilience is a challenging task, as it depends on many factors, such as risk,

Table I
Characteristics of resilience.

Four main characteristics of resilience (4R)	
Robustness	Withstanding capability: can cope with HILP events.
Redundancy	Substitute resources or alternate processes.
Rapidness	Faster approach, contingency plans, and emergency operation.
Resourcefulness	Effectiveness in handling, Prioritize.

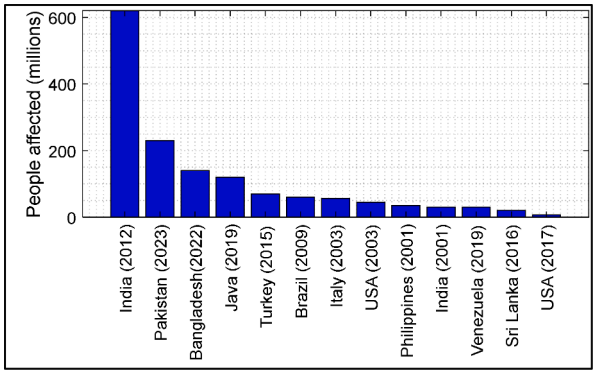


Fig. 5. Major power outages.

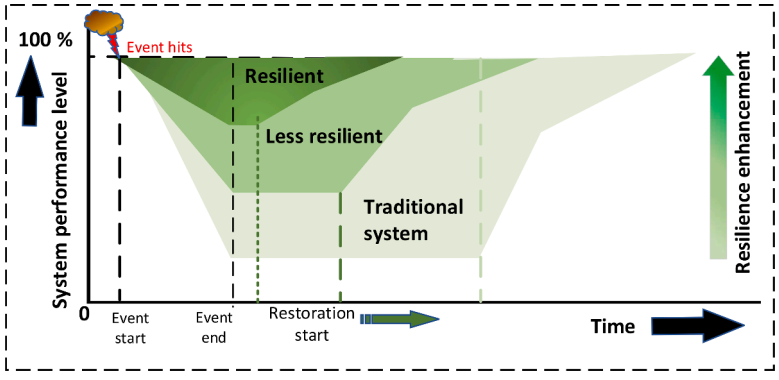


Fig. 3. Resilience trapezoid [18].

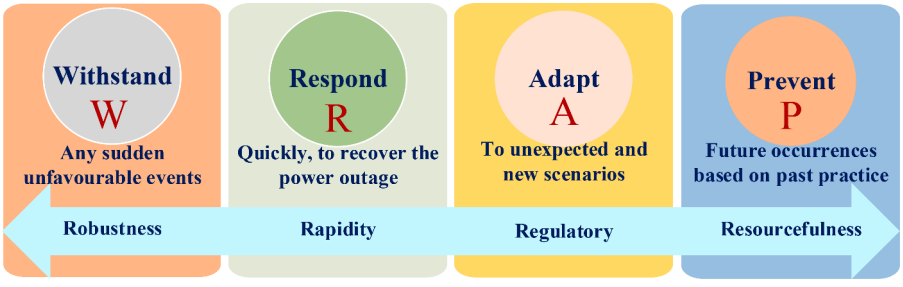


Fig. 4. Resilience attributes.

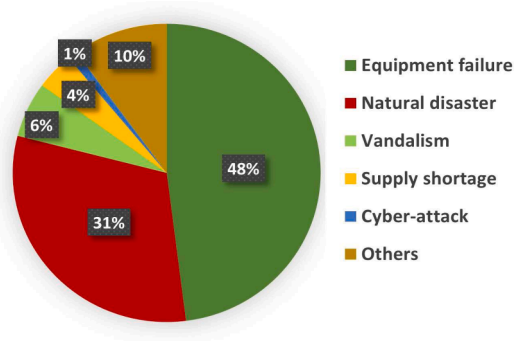


Fig. 6. Roots of power outages.

safety, and reliability. Several advanced technologies are incorporated to improve the system's resiliency [22]. During the last few decades, the frequency of extreme disruptive events has been increasing [23], which can result in the loss of robustness of electrical infrastructure. The disruptive events have several implications, such as massive economic loss and critical service disruptions, including stopping information technology-related services, water services, medical services, security, and many more [24]. Hence, several things must be considered while designing the electrical infrastructure and renovating the system robustly to withstand all possible events.

Over the past few decades, the world has seen a complete transformation of electrical infrastructure regarding reliability, sustainability, and resiliency. This insight brings to the '*light on*' ability strategy in terms of a reliable and sustainable grid, and we are now looking for a resilient grid [25]. A reliable grid defines the '*light on*' in normal operations [26]; a sustainable grid defines the grid that meets the requirements of present states without yielding the ability of the upcoming energy sectors [27]; while the resilient grid should be capable of executing preventive action, alleviating the impacts of high-disruptive events, responding effectively by automatic control method, reducing the restoration time to meet the load demand, and prospectively leading to substantial economic savings.

The contributions of the paper are summarized as follows:

- This paper opens a new perspective on power system resiliency and its urgent need against extreme natural disasters and cyber security attacks.
- An interdisciplinary study of power system resiliency is presented by categorizing it into four divisions including hardening, distributing, smartening, and building of grid infrastructures, each of which constitutes a critical pillar for sustaining the resiliency of power systems.

  A comprehensive literature review is provided by studying and prioritizing the resiliency improvement strategies for individual pillars and their connections.
- Resilience indices, measures, barriers, various challenges, and opportunities for power system resilience are discussed.

The crucial importance of resiliency and its impacts on human societies, realizing sustainable and stable smart cities for improving welfare, saving human lives, and securing businesses are clarified.

2. Resilience enhancement pillars

The enhancing strategy of power systems resilience can be categorized into four pillars, as shown in Fig. 7. These are smartening the network, hardening the structure, integrating distribution generation, and building efficiently [28–35]. Nowadays, smartening the power system network is one of the most emerging technologies to form a smart grid; it also includes automatic control of the generation and load demand, automatic fault detection, and quick restoration [36]. Hardening refers to the design of a robust structure that can withstand different operating conditions related to weather and cyber-physical attacks [37]. In addition to this, to enhance the level of resilience, distributed generation plays a significant role in electric infrastructure, which comprises renewable energy sources (RESs), combined heat and power systems, and creating more microgrids (MGs) [38,39]. The proper selection of distributed generation can enrich resiliency because the MG can be operated in an islanded mode when there is a need to isolate it from affected areas [40].

On the other hand, in building, resiliency refers to the backup system, for example, battery storage, passive solar, energy-efficient

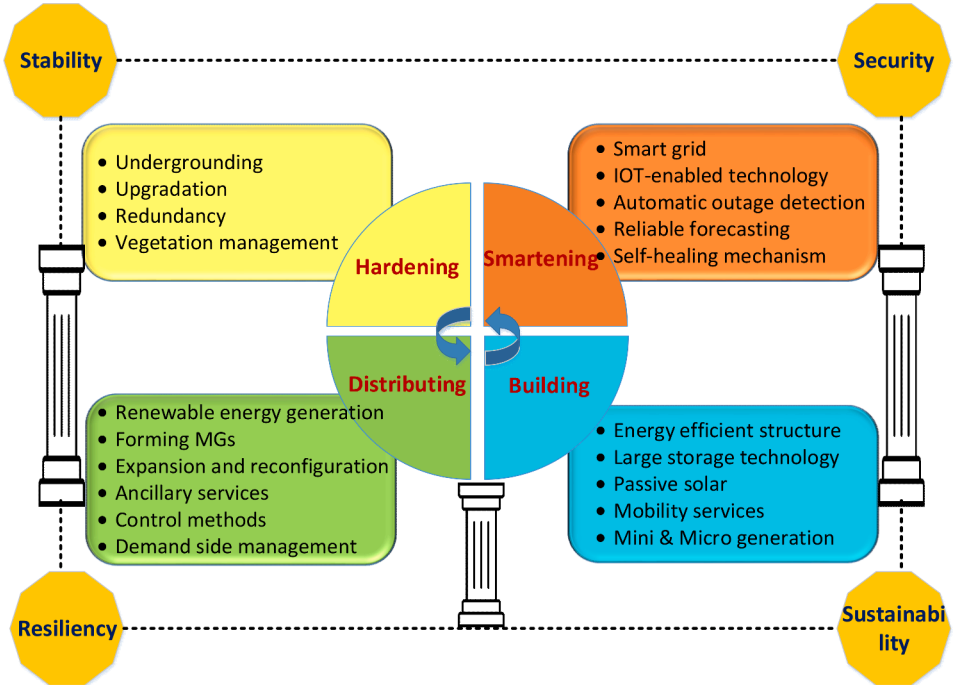


Fig. 7. Classification of Resilience Enhancement Pillars.

building, thermal mass, and natural ventilation. The detailed discussion of each enhancing pillar of power system resilience, as per Fig. 7, is as follows. Further, the conceptual frameworks for each enhancement pillar used in the available literature are presented in Tables II, III, and IV.

2.1. Hardening

Hardening measures of the resilience enhancement scheme focus on robust and resistant infrastructure [30,41]. The main objective of these measures is to minimize the physical consequence of unfavorable incidents and arrest the disability of the bulk power system network [2]. Several measures can be considered for hardenings (shown in Fig. 8), such as pole replacement, upgrading network structure, underground distribution, proper relocation of substation and transmission, where disaster consequences can meet low loss, vegetation management [35], and redundant transmission lines, are discussed below [35]. Fig. 9 portrays the three important aspects of the disaster concern: time, structure, and safety. Time denotes before and during the events. Whereas the structure must be robust to endure undisruptive events, it presents the system's survivability. Further, during the events, the response must be considered a vital factor in battle with unfavorable events; hence, preparedness is necessary. However, adjacent network coordination is important during the events to recover the system quickly, where on-site generation can play a crucial role, remarkably changing the resilience characteristics [42]. Indeed, the US Department of Energy (US DOE) has set up an action plan for hardening considering the extreme events discussed in [43]. In summary, TABLE II defines the various hardening strategies proposed by researchers to cope with disasters and eventually enhance the power system's resilience.

2.1.1. Undergrounding

More recently, literature has emerged with contradictory findings about the hardening measures with the underground distribution. For instance, in the event of an earthquake, the underground-based power system happens to have a complete loss of electricity connection, which is quite difficult to recover within a short time. In [44], the resilience characteristics of the overhead transmission line are assessed by considering the high wind speed for prioritizing the hardening support. In addition, fragility analysis is addressed to enhance system reliability. Further, the hardening of overhead lines is compared with underground pipelines in [37], and it is illustrated that the failure probability of overhead lines is 0.3, whereas that of underground pipelines is 0.1 against natural disasters. Ref. [28] reports the hardening plan of the underground system against storm surges and hurricanes, and it also presents the enhancement strategies.

Table II
Hardening strategies.

Development	Model used	Ref
Proposing reconfiguration techniques for optimal hardening strategy in the DS.	Defender-attacker-defender	[35]
Establishing a robust system hardening plan for the enhancement of resiliency in the natural gas, electricity infrastructures, and distribution grids.	Tri-level robust optimization-based	[37]
Introducing novel hardening strategies for distribution system resilience enhancement.	Two-stage stochastic mixed integer	[42]
Demonstration for resilience estimation of DS considering the strong wind hardening.	Monte Carlo simulations with fragility assessment	[44]
An upgraded tower modeling is introduced by considering the probability of damage.	Two-stage stochastic mixed integer	[45]
Improvement of the hardening strategy through redundancy is discussed.	Monte Carlo-based time series simulation	[53]
A machine learning approach is analyzed to address the power system failure	Machine learning	[60]
Narrating the enhancement of the hardening plan through vegetation management.	Pressure and release framework	[58]
A hardening strategy for large industry energy hubs is clearly presented	Stochastic mixed integer	[61]
Enhancement design of fault chain risk expectation of transmission line is developed.	Monte Carlo Simulation	[62]
A hardening strategy for large industry energy hubs is clearly presented	Stochastic mixed integer	[61]
Introduced a cooperative hardening model of critical infrastructures and pipelines.	Monte Carlo Simulation	[63]
Proposed a resilient expansion planning and hardening strategy of a multi-energy distribution framework.	mixed-integer linear programming	[64]

Table III
Distributing the resources for resilience enhancement.

Development	Model used	Ref.
A self-healing MMG model is established to enhance the resiliency of the DS.	Consensus algorithm	[38]
The role of renewable sources and conversion technologies in a smart power system is addressed.	–	[67]
A detailed review of renewable energy integration, key challenges, opportunities, and applications is presented.	–	[68]
A renewable energy-based control network is modeled and validated.	Robust grid synchronization	[71]
Standards and policies of renewable energy generation are discussed and validated.	Mathematical program with equilibrium constraints	[118]
A detailed discussion of the importance of MG in the future power system with different approaches.	–	[79]
The optimal planning model of MG is introduced for the power balance and reserve capacity operation.	Mathematical program with equilibrium constraints	[81]
Resilience enhancement models of MG are established through a multi-energy network and holistic approach.	Dynamic formation of MG	[82]
The role of interconnected MGs in the smart DS is discussed.	Blockchain technology	[86]
Different control mechanisms of MMG are presented.	Power factor-frequency inverse droop control	[91]
The risk-limiting load recovery model is developed to improve grid resilience with renewable sources.	Gaussian mixture	[119]
The load restoration technique is modeled through workforce planning with an integrated scheme.	Lyapunov-Krasovskii	[120]
The strategic developments of renewable energy integration in the DS are discussed.	Distributed control	[121]
The service restoration is done by interconnected MGs to enhance the resiliency of the distribution system.	Hybrid algorithm of Rough Set Theory	[122]
Introducing a new control method for islanded MG	Interconnected approach	[123]
Various challenges, outlooks, and benefits of the MG are reviewed.	–	[124]
An MMG model is developed with large-scale RES integration.	Hybrid Algorithm of Rough Set	[125]
The MMG management model is developed through the model predictive control method.	Distributed economic model predictive control	[126]

2.1.2. Upgradation

Upgradation is one of the most significant factors for hardening. Certain things can be considered in upgrading, such as upgrading poles and towers, elevating substations, and using technology advancement. In [45], upgrade modeling is done against the probability of damage. Various case studies with different constraints are also considered. Fragility modeling of the transmission line is an important part of the upgrade, which can measure the failure rate irrespective of the wind speed intensity, as seen in [46]. On the other hand, elevating substations

Table IV
Smartening methods of the power system.

Developed	Model used	Ref.
Resilient and optimal operation improvement method of DC MG through the hierarchical control method.	Hierarchical control strategy	[96]
The resiliency of the system is improved by attack detection and alleviation using the AGC model.	Attack resilient control	[97]
Enhancing the transient stability and resiliency of the power system through the distributed control scheme.	Distributed nonlinear robust controller	[128]
Improvement of the transient stability in the multi-agent system through governor control and energy storage system.	Governor-based power control	[129]
Overview of power system deficiency and the importance of smart meter in the smart grid.	–	[131]
The potential benefits of blockchain in the power system to enhance smart grid resilience.	–	[132]
Resiliency enhancement methods to restore the operation through distributed energy resource integration.	Alternating direction method of multipliers	[134]
The application of ST in an MG to control the voltage for improving the system resiliency.	Voltage-based droop control	[136]
The use of smart transformer in an SG for automatically identifying the faults and condition monitoring of the power transformer.	Signal processing and pattern recognition	[137]
To increase the survivability of the distribution system, the solid-state transformer (SST) based MG and multi-microgrid model are presented.	SST integration	[138]
Enhanced voltage protection and frequency regulation schemes using SST in the distribution system are introduced.	SST integration	[140]
Proposing load forecasting techniques using machine learning against cyber-attack.	Deep learning framework	[156]
Integration of renewable energy through SST to enhance the resiliency and automatically control the loads and other enabling technologies are discussed.	SST integration	[165]
A defense strategy for resilience enhancement using islanding techniques is discussed.	Tri-level robust optimization	[166]
The frequency regulation of MG is enhanced through the electric springs.	Frequency and voltage control	[167]
An improvement strategy of frequency regulation and controlled islanding is introduced.	Mixed integer linear programming	[168]

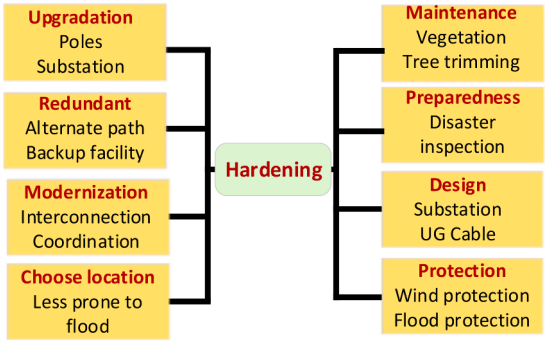


Fig. 8. Hardening module.

is another approach to hardening the electric infrastructure. A few studies have been carried out in this area, as seen in [47,48]. In [49], substation elevation is discussed with their cost-benefit analysis using the Markov process. A preventive flood damage-based substation model is presented in [47], where Harvey's 100-year rainfall is taken as real data and is regarded as the mitigation strategy under the hardening plan.

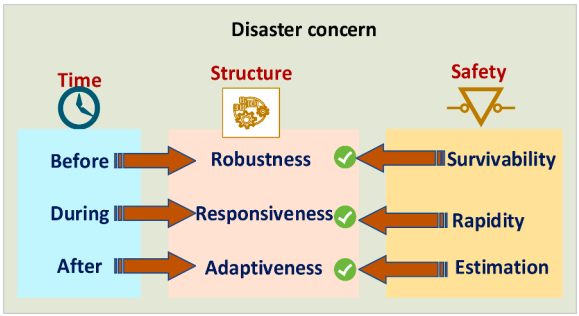


Fig. 9. Hardening response.

Ref. [25] presents the day-ahead hardening plan through the substation protection using crew team scheduling. In addition, relocation of a substation is an important concern regarding the less prone to flooding. Many customers will be in trouble if the substation is frequently affected by bad weather or flood conditions. Thus, the substation relocation can prevent the power outage [50]. Indeed, technological advancement is always a better achievement plan for hardening [51].

2.1.3. *Redundancy*

Some particular transmission paths could face power disruption issues in the wake of extreme events such as hurricanes and wind storms. So, designing redundant transmission routes helps minimize power disruption, reducing the effect on customers and increasing resiliency [52]. Many studies have demonstrated resilience enhancement using redundant transmission paths, which can be seen in [52–54]. In [53], the authors illustrate redundancy using various types of networks, such as normal, robust, redundant, response, and recovery and then measure the loss of load index. On the other hand, the self-healing defense strategy of power systems against cyber-attacks is presented in [54], and resilience enhancement is shown using redundancy. To make the grid stronger, bigger, and smarter, redundancy plays an important role, which is the critical resilience feature, and the details can be seen in [55].

2.1.4. *Vegetation management*

Vegetation management is a preventive hardening plan considering tree trimming surrounding the transmission lines, which can help to reduce the flying debris that contacts the lines, resulting in short circuits. The distance between lines and trees can be maintained through tree trimming, which also helps crews access the area for repair without risk [56]. It is indeed important for vegetation management to maintain the clearance between the tree and the tower. As noted, 90% of power outages in the United States are due to distribution system (DS) issues, where improper vegetation management is part of the reasons [57]. Thus, proper vegetation management is crucial to reduce power outages and increase resiliency. The authors in [4] precisely discuss the importance of vegetation management to a power outage as well as the corresponding resilience. Ref. [58] narrates the physical vulnerabilities of power systems due to vegetation management. On the other hand, the dynamic model and framework for vegetation management are developed, and the effectiveness against different outage conditions is presented in [59].

2.2. *Distributing*

Distributing is a crucial part of a resilient power system. This can help build a resilient power system through various planning and operational schemes, such as incorporating renewable energies, restructuring ancillary services, and adopting control algorithms. This scheme can help when a disaster hits the network and it experiences a major blackout, which leads to damage to the power system. Therefore,

it is most important to have a quick action plan and operation using resource allocation in the power system [65]. The allocated resources are used to restore the failure of emergency load in the power network. The important aspects of distribution in the power system context are discussed below and summarized in TABLE III.

2.2.1. Renewable energy integration

Power producers emphasize renewable energy to meet the NetZero target and the increasing power demand. It can be predicted that renewable sources, like wind, solar, and biogas, can be considered for supplying power as per the demand [66]. There is an extreme diversity of loads exhibited in smart modern grids, along with the demands of efficient consumption [67]. The growth of renewable energy usage, coupled with grid connection technology, brings about the challenges and problems that have been faced while governing power quality. This has been accomplished through the interfaces of power electronics [68]. Compared with traditional power generation systems, power electronics interfaced systems have more efficient energy conversion methods [69]. On the other hand, the integration of power electronics into the power system poses great challenges, particularly influencing grid stability [70]. These challenges are complex and non-linear when the power system is integrated and intermixed with thermos, electro-dynamics, and chemical energy [71]. In addition, due to generation distortions, connecting traditional converters directly to the grid is difficult as the output voltage is high, and the single switch cannot manage it at the grid's voltage level [72]. Notably, inertia is critically important for maintaining system stability as it supports dampening oscillations. To tackle the stability challenges, the power system must be equipped with advanced control strategies. Mehrasa et al. proposed a single synchronous controller to control the inertia that has been integrated with renewable energy sources [73]. In addition, a few other researchers have also studied the control of grid-connected converter using various control techniques, such as synchronous power controller [74], self-synchronized synchronverter [75], synchronous conventional droop-based control [76], virtual synchronous machine [77]. With these approaches, renewable penetrations are demanding, contributing to the enhancement of stability and resiliency in the power system.

2.2.2. Microgrid

MGs are future power systems that provide clean and economic power to the utility network [78]. In the case of an MG, energy sharing between producers and consumers is a point-to-point system combined as an isolated energy system [79]. MGs are the main foundations of smart grids, which enable a self-controllable power system through distributed energy-based resources and variable loads (can be seen in Fig. 11 as an isolated microgrid (IMG). These isolated power sources have been combined to cater to the energy needs of standalone communities. In case there is a surplus of energy generation, MGs become self-sufficient and work independently. Therefore, an MG is essentially considered a small power system consisting of several isolated power-generating units [80]. MG configuration needs adequate control and management, which provides many more advantages to the beneficiaries, such as more reliability, flexibility, and better power quality combined with less generation cost. Furthermore, if surplus energy is generated from the MG, which shows the energy involved is not equal to the energy consumed by the individuals, the remaining energy will be sold to a utility company or stored in local storage systems; similarly, when there is a shortage, the energy can be bought from the utility company [81]. Regarding resilience, the MG system has better restoration capabilities in response to natural disasters because it can operate in either grid-connected or islanded mode [82]. Indeed, resilience improvement requires faster and more effective control strategies to manage outages efficiently. These requirements can be fulfilled through the integration of all autonomous MGs. Besides, the system vulnerability can also be reduced, which leads to saving the damage cost [83]. Given all that has been stated so far, one may understand that the installation of an MG is a better solution for reliability and resiliency. In recent years, there has been a growing interest among researchers in interconnected MG rather than single MG, which is further explained as follows.

2.2.3. Expansions and reconfiguration of networks

Nowadays, the decentralized power system is increasing worldwide, forming MGs and further extending them to become multi-microgrids (MMGs). MMGs are occasionally called interconnected MGs or networked MGs that can be stated as a couple of MGs organized in extensional closeness with a corresponding energy management system for collaborative support and power exchange, as shown in Fig. 10. Over the past few years, increasing attention has been observed from single MG to MMG for enhancing the power system resiliency. On the other hand, a multi-layer MG system is developed to operate the MMG system smoothly, considering the reliability and resilience factor to cope with high-disruptive events. Indeed, each system layer has its own structure and is monitored by the service layer [84]. Further, in [85], a self-governing operation of coupled MGs with AC common bus is pioneered. For smart distributions, coupled MGs through utility feeders are discussed in [86], and DC link-based robust interconnected MGs are presented [87]. Furthermore, island-based MGs are presented, focusing on a coordinated dispatch scheme for power balance under different disruptive events [88]. The application of MMGs enhances the automated control of the DS under blackouts and is demonstrated in [89]. The self-healing strategy of MG operation can substantially improve resiliency economically in the wake of extreme events, as discussed in [38]. Further, a detailed discussion of the system operator and MG and its energy-trading strategy is reported in [90]. The preceding studies provide evidence that the coordination of networked MGs is a large time-scale operation. However, due to practical constraints, MGs cannot provide the exact feasible solution for a reliable power system. Thus, rapid power exchange during transient stability operation through interconnected MGs, which is the resilience enhancement scheme of the power distribution system, can be suggested.

The most important distinguishing feature of MMGs is the rapid power-sharing capabilities among the interconnected distributed generators. However, a standalone MG can also be attained in tandem through voltage or frequency control in a centralized or decentralized manner [91]. More recent literature has motivated the extension of the single MG to MMGs to prevent power disruptions and diminish communication overhead [92]. One of the major concerns about the MMG in relation to resiliency is that each MG is connected through a tie-line or link so that the disaster-affected area can be easily isolated from a healthy area. Further, it will be less likely to affect the customers. Besides, MMG has a better energy management capability to battle against unfavorable events, enhancing the power system's resiliency [39]. It is important to note that the MMG formation can transform the power system into a resilient electric infrastructure to maintain dynamic stability, critical load supply during disasters, and quick restoration, thus subsequently decreasing the recovery cost. In addition, appropriate repair crew scheduling and EV infrastructure could help the power system load restoration after the events, minimizing the consequences and enhancing the resiliency, as discussed in [93,94].

2.2.4. Control methods

The power system is a complex network where various operating states are defined; these states depend on the parameter values, such as voltage, frequency, and power. Inaccurate control can destabilize the power system network, leading to power outages [95]. Indeed, the power system can be largely affected when it faces cyber threats, and its defensive mechanism can be designed through a control unit. Recently, decentralized power systems have increased, which forms the MGs where various control methods, such as primary, secondary, hierarchy, and distributed control, have been adopted to cope with faults and unforeseen events. A hierarchical and distributed control-based MG is presented in [96], where resilience enhancement is shown. Notably, an

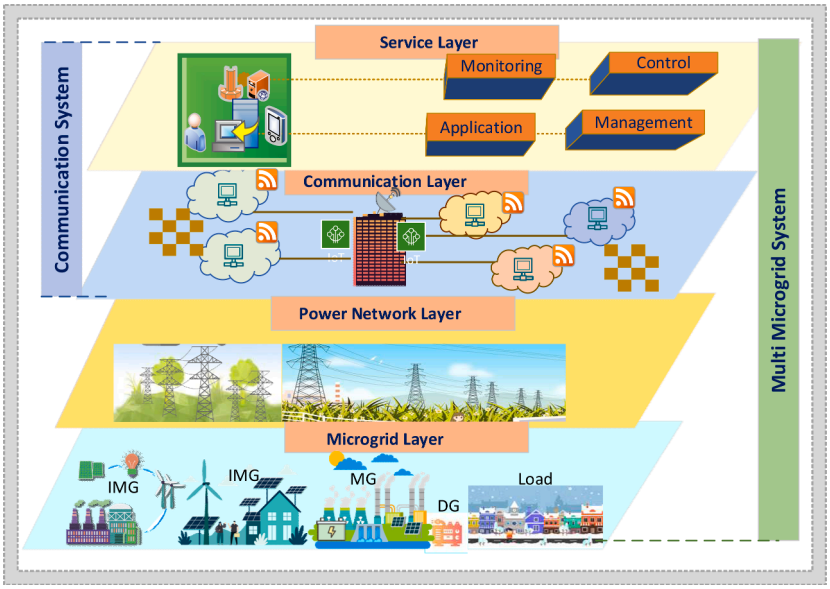


Fig. 10. Multi-layer-based MG systems for achieving resilience.

important control unit of a power system is automatic generation control (AGC), which has tremendous importance concerning the power system's stability.

The role of AGC in the power system is vital which is to uphold the frequency and tie-line power in a permissible range. With this objective attained, the area control error (ACE) will be calculated through supervisory control and data acquisition (SCADA) every few seconds, and then it will automatically control the frequency and exchange of power. The values of ACE specify the balance and imbalance between load demand and generation, which is to be regulated through the governing system. The integral requirement of control and the measurement records on wide-area SCADA communication makes the AGC program more vulnerable to various kinds of cyber-physical attacks, which could eventually influence the system frequency.

Over the past few years, cyber-attacks have been increasing, and various data integrity attack models have been established [97]. Besides, the attack resilient control model is developed to identify and alleviate attacks from different sources [98]. Furthermore, to enhance the AGC stability, accurate load forecasting is necessary, and malicious data interpretation has to be minimized. The detection and mitigation technique of malicious data from real-time forecasting is developed and presented in [99]. It also can stabilize the power system operation and extend to a resilient AGC system. Figs. 11 and 12 portray the AGC attack-resilient model and AGC wide-area monitoring, regulatory, and protection-based model, respectively. The symbols ACE, f and P_{tie} denote the area control error, frequency, and tie-line power, respectively.

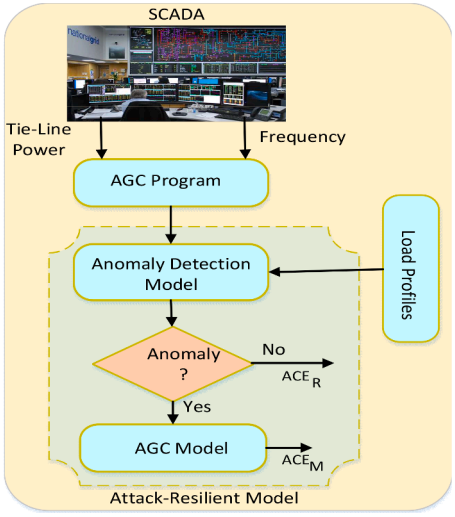


Fig. 11. AGC attack-resilient model.

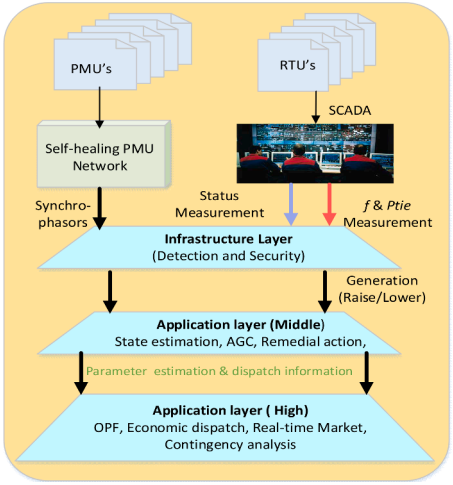


Fig. 12. AGC wide-area controlled structure.

2.2.5. Ancillary services

The ancillary services generally maintain the power system's stable operation, which further realizes the system's security, reliability, and resiliency. These services mainly deal with generation reserve, frequency and voltage control, active/reactive power support, etc., as illustrated in Fig. 13. As far as the generation reserve is concerned, a variety of resources are used, such as hydro-pumped storage [100], battery energy storage [101], electric vehicles [102], and hydrogen electrolyzers [103]. A novel resiliency scheme is discussed through ancillary service using island operation capability, which has been realized using distributed energy resources [104]. The ancillary service is of utmost importance in frequency regulation, as a large frequency deviation can lead to a blackout. In [105], a power disruption mitigation scheme, under-frequency load shedding, is presented using ancillary

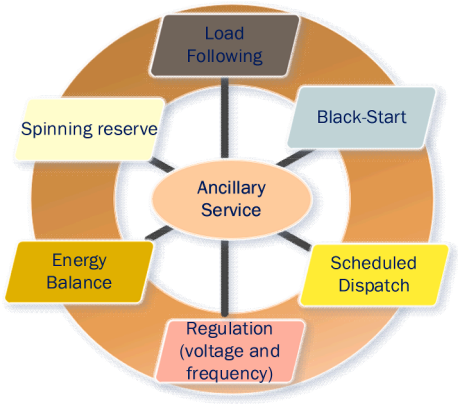


Fig. 13. Ancillary service capability.

services to make the system resilient. Further, a supercapacitor as an ancillary service is used to enhance the stability of the power system, as presented in [106]. In the decentralized power system, the reactive power balance is needed to stabilize the power system, and for that, a few ancillary devices are used to compensate for it; details can be seen in [107].

2.2.6. *Demand side management*

Demand side management (DSM) is the key aspect of the modern power system, which allows customers to optimize their load to balance the energy during peak time, resulting in economic benefits [108] and a few more capabilities, as shown in Fig. 14. On the other hand, incorporating DSM into the smart grid can improve reliability, overall efficiency, and resiliency [109]. While facing high-disruptive events, the grid might not be able to meet the total electricity demand, but it can meet the partial demand, which is inherently risky in case of a power outage in critical infrastructure [110]. To reduce the risk, DSM plays a vital role in managing the load scheduling as per the priority level [111]. In the last few years, several efforts have been made to show that the DSM is key to improving the grid's efficiency and resiliency [112–114]. A few recent studies on DSM demonstrate that this approach is the best fit for MG and MMG systems, which has been evidenced in terms of cost-effectiveness [115], low-carbon transition [116], and resilient operation [117].

2.3. *Smartening*

In the last decade, the energy sectors have undergone a significant transformation by incorporating advanced technologies, metering services, communications protocols, a digital platform, and interconnected services. This shift has led traditional power systems to evolve into smart

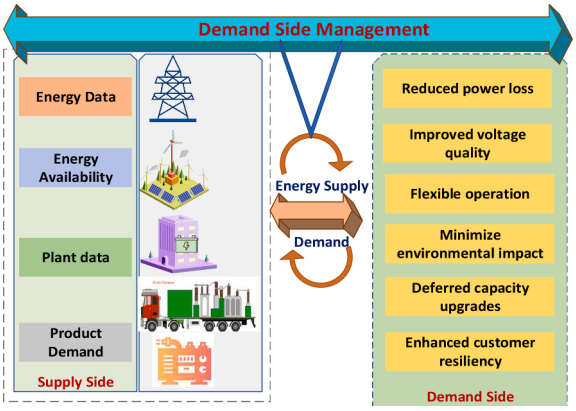


Fig. 14. Demand side management capability.

power networks, enabling reliable communication. In addition, enhancing the system's reliability, security, and resiliency. The various approaches to smartening the power system are discussed below, and the researcher's contribution summary can be seen in TABLE IV.

2.3.1. *Smart grid and its realization*

Smart grid technology is an emerging area of the power system network that can improve grid resilience. As noted, resilience enhancement is vital to all critical services to minimize life threats [127]. With this objective, the smart grid is one solution that intelligently uses advanced technologies to manage the electricity service to users in a reliable and resilient manner [22]. Several research studies have been conducted to elucidate the smart grid-based resilience enhancement strategy. Ref. [128] has demonstrated the transient stability enhancement scheme of the smart grid considering the resilience factor. Besides, a distributed non-linear controller in association with communication delay and cyber-physical perturbation is proposed in [128]. A governor control-based power system with storage units is discussed in [129], where the proposed method is validated using the IEEE-68 bus system with better stability characteristics. Information and communication technology (ICT) plays a significant role in smart grid operations. The benefits of ICT on account of cost, resilience, quality, and security topologies, irrespective of sensitive loads, are discussed in [130]. Further increases in power line threats lead to the mismatch of load and generation, which causes the frequency to deviate from the nominal value, and thus, a power outage may happen. In order to reduce power outages, the smart meter can be deployed on the prosumer end so that the customer load profile information can be seen, decreasing data loss and instability [131]. The blockchain database is developed in [132], where all the secured data can be stored, enhancing the smart grid cyber resiliency and safe power system transactions. There is another growing body of works discussed regarding various strategies to increase power system resilience, which are presented in [133]. In [96], the economic operation of DC MG is discussed through a hierarchical control strategy. The proactive operation strategy towards the enhancement of power system resiliency is presented in [117]. Further, the risk-limiting restoration technologies are presented in [134], where quick restoration methods are mainly focused on through the integration of RESs.

On the other hand, in the smart grid (SG), the role of smart devices, such as smart transformers (STs) and smart meters, is crucial. The ST deals with voltage swell/sag mitigation, reactive power compensation, and harmonic distortion and protection [135–138]. In addition, the ST operates very fast to control the input/output signal during the fault condition. In [138], the authors provide the most extensive noteworthy applications of ST in the power grid to improve the system's resilience. As noted, several MGs can be established through ST, from a large traditional connected system to the small unit operating in an islanded mode in the wake of an unforeseen event. The fundamental objective of this design is to achieve robust and resilient operations [139]. Furthermore, these days, power electronics devices are capable of developing intelligent and smart devices like STs, which are a piece of cost-effective and functional equipment superior to the traditional transformer and have excellent isolation properties. It is the flexible interface between low-voltage and medium-voltage power systems and can significantly increase the power system resiliency [140]. On the other hand, smart metering is the primary initiative for fulfilling the aim of SG. It solves critical issues such as load demand management, customer operations, energy balancing, etc. The important thing about a smart meter is that it stores data that can be further used to communicate with the system operator to balance the energy, reduce energy theft, and increase energy efficiency. Regarding resilience, smart meters, end-to-end privacy protection, and a few other security schemes are used to defend against cyber-attacks [141]. Several studies have revealed that smart metering significantly contributes to smart grid resilience in various ways, as discussed in [142,143].

2.3.2. Internet of things enabling technology

In the fourth industrial revolution, data-driven technology is important due to the increase in the Internet of Things (IoT) [144]. This significantly benefits the energy transition using digital technology. Further, the IoT-enabled platform offers a reliable, efficient, secure, and resilient power system as it deals with real-time monitoring, smart control methods, situational awareness, and cyber threat protection [145]. Nowadays, IoT technology is embedded with the advanced metering framework and SCADA to develop intelligent power systems [146]. Several research studies have reported IoT-based power systems in [147,148]. Moreover, the IoT in power systems enhances the operation capabilities, home appliances control, end-to-end facility provisioning, enhanced scalability, minimizes physical attacks, and finally helps to improve resiliency.

2.3.3. Automatic outage detection

In the context of power system resilience, preparing for detecting the outage locations and mitigation strategies are highly significant. An efficient outage detection can greatly reduce the customer-affected time due to extreme events. As noted, manually finding faults or outages in the power system is an intensive, complex, and prolonged process. SCADA offers operational information at the lower level, such as the distributed level. In the past few years, advanced metering has been incorporated to provide further clarity of outage detection [149]. However, advancing metering (AM)-based techniques need a centralized system to detect the information, which is not cost-effective and unreliable. Thus, to provide better detection techniques and restoration, a few authors have demonstrated using various restoration algorithms through the reconfiguration method [150]. In [151], an AM-based power outage detection method is presented, and MMGs do the restoration. A malicious data detection-based study is conducted in [152] to show the resilience of the smart grid. Artificial intelligence techniques have recently been applied to outage detection problems to enhance the power system's resilience [153,154].

2.3.4. Reliable forecasting

Reliable forecasting can play a critical role in power systems, decreasing vulnerability in the face of natural disasters and cyber-attacks. During the past decade, there has been a significant development in forecasting techniques using sensor measurements [155]. On the other hand, various forecasting models have been used to predict the required parameters to evade power system failure. Indeed, load forecasting and weather forecasting are the most used strategies in the power system. Refs. [156] introduce load forecasting techniques against cyber-attacks, which apply the artificial intelligence method. Further, a novel predictive power system is designed against high winds, and resilience is also measured, as can be seen in [157]. A Markov-chain-based forecasting technique is reported in [158], where the authors present the operational efficiency and resiliency of the power system. Considering the power system stability due to intermittent resources, artificial intelligence techniques-based forecasting is developed in [159], which enhances the system's resilience performance.

2.3.5. Self-healing mechanism

An important part of power system resilience is a self-healing mechanism against extreme events. After extreme events, the first and foremost thing is a defensive action using the available resources to avoid cascading effects. Various defensive strategies can be adopted, such as islanding and reconfiguration, which can be done by using various communication channels, sensors, and control methods. With these technologies, real-time data monitoring can be done, and the network's troubles can be found. Thereafter, defensive strategies emerge to evade power outages and enhance the system's resiliency. A substantial number of studies have developed the defensive mechanism through islanding techniques [160,161]. Further, other self-healing strategies to cope with unforeseen events are multi-agent MG [38], load shedding [162], phasor measurement unit network [54], reconfiguration restoration [163], and protection system [164]. Moreover, the self-healing ability of the power system aims to reduce the grid vulnerability and restore the operation automatically after unforeseen events, which is a distinguishing characteristic of the smart grid.

2.4. Building

In recent years, the increasing frequency and severity of extreme events have been high; thus, building a resilient power system is one of the key necessities [55], which needs to be accomplished through various improvisation paradigms, for instance, energy-efficient buildings, energy storage backup facilities, passive solar technologies, and thermal mass and natural ventilation arrangements. Using these techniques, the power system is not only referred to as resilient but also proved to be an efficient and better operative performer during emergencies and disruptive states [169]. The various methods of building resilience on demand are described below, and a brief contribution is presented in TABLE V.

2.4.1. Energy-efficient structure

The policies related to energy management are trying to deal with the problems involved with energy across the globe. The policies are energy efficiency, energy crisis, and energy resiliency, which concern environmental issues, climate change, and infrastructure. These issues can deal with short-term and long-term energy management plans to build a sustainable framework [170]. The author in [171] reviews the various approaches for energy-efficient building plans in terms of their performance through architectural shapes. In the past few years, developed countries have been building green homes to achieve the zero-carbon target and, eventually, have realized improved energy efficiency and resiliency [172]. The climate-resilient infrastructure model is introduced in [173], where energy efficiency is shown. Refs. [174,175] promote building an energy-efficient model considering resiliency in an urban-scale infrastructure. A model predictive control-based energy-efficient building is narrated in [176], where authors consider the weather forecast to control the temperature. Further, a detailed

Table V
Building strategies for power system resilience enhancement.

Development	Model used	Ref.
The energy-efficient building model through passive solar design with comfort zones and control potential zones is presented.	Passive climate control	[190]
The rural application of the passive solar model and its collecting methods are addressed.	Photothermal conversion	[193]
The various examples of passive solar implementation in different countries are presented.	Overview	[191,194]
The optimal location of the energy storage-based models is discussed on account of the power system resilience enhancement.	Non-black-start with optimal sizing	[197]
Various approaches to mobility services to enhance resiliency are presented.	Overview	[198,199]
A VPP-based electric infrastructure is modeled and presented.	Distributed control	[209]
The role of resilience in electricity and its comparison are presented.	Overview	[211]
An overview plan of changing global policy towards the energy-efficient structure for enhancing the resiliency of power systems.	Overview	[212]
A proper energy building and management strategy according to its specification and consumption is discussed.	Case study	[213]
An important discussion of building resilience electric infrastructure	Overview	[214–217]

discussion of energy efficiency and energy resilience is presented in [177]. Notably, it is clear that a better energy policy in regard to efficient building as infrastructure can cope with extreme events.

2.4.2. Storage technology

Electrical energy storage (EES) units are essential for DSs to support the grid during an emergency and enhance power system resilience [178]. Conventional methods of EES units mostly emphasize reliability necessities instead of resilience. For dealing with both reliability and resiliency, sizing and optimal allocation of EES units should be considered. Moreover, EES units can attain significant assistance, like enabling renewable technologies, minimizing operating costs during normal operation, enhancing the power system reliability in distinctive fault scenarios, and improving resiliency under natural disasters [179]. Various types of operative scenarios in power system networks, with EES units considering reliability and resilience, are presented in [180]. Due to the rapid development of storage technology, large storage batteries have entered the market, which are now used in the power system to cope with extreme events. The application and challenges of energy storage using renewable energy sources on a grid-scale can be seen in [181]. The size of the storage units in terms of pricing and energy management is discussed in [182]. The initiation of the world's largest storage technology in operation is presented in [183]. The impact of energy storage on the power system's resilience is illustrated in [184]. In recent days, the battery capacity has been increased significantly to support the load during emergency conditions; for example, Vistra's Moss Landing energy storage, US (1600 MWh), Manatee energy storage center project, US (900 MWh), McCoy solar energy project BESS, US (920 MWh), Elkhorn battery, US (730 MWh), and Victorian big battery, Australia (450 MWh).

Energy storage technology has grown significantly in the last few years, and many breakthrough technologies have been established. One of the revolutionary technologies that have been developed [185] is the solid-state battery (SSB), which has excellent benefit characteristics as compared to existing batteries, such as safety enhancement, higher energy density, faster charging/discharging, longer life span, operate wide ranges of temperature, compact design, reduced environmental impact, and lower self-discharge rates. On the other hand, the incorporation of SSB into power systems can have a significant impact on improving resilience, and it can contribute to robust electric infrastructure [186]. The contributions of SSB to the power system resilience are providing higher energy density to enhance the stability of the grid, fast response to fluctuation, less dependency on the backup as it can balance the grid, black-start capability, maintaining stable voltage and frequency levels, can cope with high ranges of temperature. However, some challenges remain to overcome, such as manufacturing scalability, cost minimization, and feasibility with all operating conditions [187].

2.4.3. Passive solar

Over the past century, there has been an increased interest in energy-efficient buildings using passive solar techniques. It is now well established that passive solar is referred to as breakthrough technology to make a home or office into an energy-efficient building through solar heating and cooling designs [188]. Besides this, it is important to mention the energy gap because it is not directly fed from the power grid or standalone mode of energy sources. However, this technology can be installed on a rooftop solar system with proper design standards. Moreover, specific design requirements can be met with appropriate sizing of efficient windows and doors to respond to natural ventilation and lighting [189]. The realization of this design has a great impact on an energy-efficient concept system, which leads to enhancing the power system's resiliency. Further, it minimizes the power system vulnerability and energy costs. This breakthrough technology is termed psychophysics [190]. Significant analysis and discussion on passive solar are discussed in [191], and a comprehensive review is presented in [192]. A detailed study of the design and implementation of passive solar technology installed in China for residential purposes is shown in [193]. The solar-thermal performance analysis is evaluated in a passive solar building in the Republic of Korea, presented in [194]. Similarly, a passive solar heating and cooling system is installed and tested in Rajasthan, India [191]. Further, a new green home can be a solution with the development of passive solar techniques initiated by the Australian government [195].

2.4.4. Mobility services

Recent trends in restoration techniques in power systems originate the proliferation of mobilization methods after extreme events. One is that energy storage support has led to an expansion of studies that introduces a new developmental facility to provide power during an emergency through mobile energy storage units [196]. This facility will restore the power outage quickly and protect the grid from vulnerability in response to disasters. Additionally, it provides the "light on" services to critical infrastructures, such as hospitals and communication network offices. Along with these benefits, employing mobile ES units in the power system can reduce load shedding, reduce recovery time, and improve resiliency [197]. A few studies in [18,198,199] establish the distribution system model with mobile ES units to show resiliency in recovery time. On the other hand, in the last few years, various physical innovations have been developed to drive electric vehicles (EVs) successfully, which can significantly enhance resilience [102]. Recent studies have illustrated the importance of EVs in electric infrastructures and their contribution to resilience [200,201]. In addition to mobile energy storage, there has been notable development of mobile EV charging infrastructure in recent years. This development holds excellent potential for improving resilience, especially considering the vulnerability of fixed EV charging infrastructure during severe weather events [202].

On the other hand, the substation fails to provide the power supply due to extreme events such as windstorms, ice storms, and equipment failures. During these eventualities, mobile substations and mobile MG can significantly help the power system network meet the load demand. A comprehensive model of the mobile substation and mobile MG are discussed in [203,204]. Further, transmission line faults or other external faults can substantially affect the power system recovery after the events. This can be reduced by allocating a group of crew members so that they can mobilize as per the fault locations, and finally, restoration can be done faster. This is sometimes called workforce planning; it is related to the "right" repair crews in the "right" moment and "right" location to achieve organizational goals [205]. Fig. 15 depicts the mobile services used in electric infrastructure to speed up the recovery process after extreme events.

2.4.5. Mini and micro-generation

Recently, the energy transition has been adopted favorably, where the formation of a decentralized power system is a key factor. As far as the decentralized power system is concerned, it is mainly at the distribution level, where mini and micro-generation greatly help to achieve the principal objective of the energy transition, energy resilience, and zero-carbon target. Mini and micro-generation usually supply energy to remote areas. This not only provides the remote area power supply but can also connect with a larger system as a hybrid approach. It can be operated as an isolated MG system during grid failure conditions. Doing so can provide the critical service supply requirement to minimize the effect after the disasters. A substantial amount of research has been carried out in this area to show the importance of mini and micro-generation to power system stability and resiliency [206,207]. The mushrooming of mini and micro-generation technologies has emerged as an innovative concept to build a cost-effective and resilient system named virtual power plant (VPP). The authors in [208] discuss the modeling of VPP using renewables in a decentralized way. The resilient strategy of VPP in the face of cyber-attacks is reported in [209], and the optimal regulation of VPP is presented in [210]. In addition, a proper

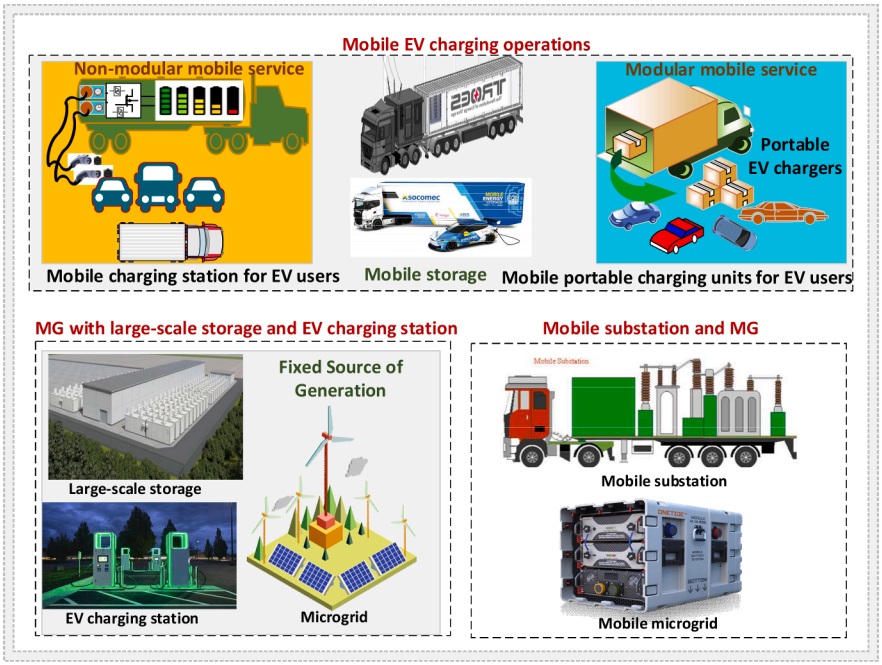


Fig. 15. Mobile resources to enhance the power system resilience.

energy building and management plan according to its specification and consumption is discussed in [211].

3. Resilience quantification, measures, and analysis

3.1. Resilience metrics

The power system comprises three subsystems: generation, transmission, and distribution. Different extreme events, such as weather-related, cyber-attacks, physical, and cyber-physical, can impact all these systems. Numerous research studies have been carried out in power systems considering extreme events and developed various resilient-based designs with quantification indices, which are discussed in TABLE VI.

In addition to the above, numerous studies have introduced resilience indices in various ways following extreme events. These include consideration of power or energy recovery, time index, frequency metrics, probability matrices, and curve matrices [241]. In terms of resilience, the power and energy recovery indices are measured by unserved load [55], critical load recovery [29,242], energy recovery [243], connected and unconnected generation capacity [30,244], and expected energy not supplied [245], Customer Average Interruption Duration Index (CAIDI) and System Average Interruption Duration Index (SAIDI) [246]. Regarding the time index, load curtailment duration [247], Loss of Load Expectation (LOLE) [248], pre/post-event degradation state [14, 30], and recovery duration [249] are considered resilience measures. Further, the frequency index can be estimated in various ways, such as System Average Interruption Frequency Index (SAIFI) [246] and outage customers [160,250]. Probability matrices are measured considering resiliency: Loss of Load Probability (LOLP) [251], probability of restoration [2], supply interruption [252], and system failure [249].

On the other hand, several other factors must be measured when addressing extreme events, such as economic, social, geographic, safety, and health, as can be seen in Fig. 16. As noted, power system damages following extreme events incur astronomically high costs, often amounting to billions of dollars [253]. A few researchers have studied resiliency in terms of damage cost using conditional value-at-risk (CVaR) [254], operation cost [255], unsupplied energy cost [256], and regional gross output [16]. In addition, the Value of Loss of Load (VOLL) is also an important factor in measuring the outage cost, as discussed in [257]. Further, power system damage cost due to extreme events can be estimated using Lost Revenue Impact (LRI), introduced in [233]. Concerning the social factors, resilience measurement has been established using customers affected by disasters, which can be seen in [242,258], communication failure [259], and critical infrastructure resilience [260]. Geographic location is an important aspect of resiliency that has to be taken into account considering the extreme events, and a few studies have considered this, such as islanding mode [35], mobile units [247,261], and geographic information system (GIS) based restoration [262]. Particularly, GIS technology could significantly help determine the disaster response per the location preference and number of customers. Finally, safety is always an important concern considering extreme events, and to deal with that, an appropriate response should be prepared, such as critical load recovery, as it is more of a priority to minimize the life threats [263], pre/post-disaster preparedness [264], and backup units like diesel generators, energy storage [265]. Furthermore, a systematic life-cycle assessment of the power plant is crucial to prevent component failure, reduce power system damage, and optimize overall performance [266]. This can be achieved by periodic maintenance schedules [267], aged infrastructure replacement [268], degradation analysis, and adaptive control implementation [269]. Therefore, technology and infrastructure upgradation are equally important to ensure the power system remains stable, sustainable and resilient with present and future challenges in the face of extreme events [2].

3.2. Resilience enhancement measures

Enhancing power system resilience is vital to combating extreme events. The power system operates on three different levels—generation, transmission, and distribution—and each level is crucial in addressing these events [270]. These measures not only keep the lights on for end-users but also defend critical infrastructure, ensure minimal power system damage and threats to life, and contribute to building a sustainable future. At the generation level, protecting the plant, whether a conventional or distributed generation-based plant, is vital. Integrating energy storage into the generation level is critical to withstand insufficient generation or face extreme events [233]. An integrated approach has been implemented in the last few decades, connecting

Table VI
Resilience metrics.

Level	Events	Metrics	Ref.
Generation	Earthquake	Developed a resilience index based on failure of probability corresponding to peak ground acceleration.	[218]
	Extreme heat conditions	Proposed temperature-dependent capacity variation as resilience index.	[219]
	Extreme heat conditions	The temperature-response function index is proposed as a resilience metric.	[220]
	Extreme heat conditions	A Quantitative vulnerability index is measured.	[221]
	Extreme heat conditions	A climate change influence index is proposed as a resilience index.	[222]
	High wind storm	The stress concentration factor is considered a resilience index.	[223]
	High wind storm	The probability of turbine tower buckling is considered a resilience index.	[224]
	Flood	Total effects index of damage	[225]
Transmission	Extreme disaster	The system resilience index and component resilience index are introduced.	[226]
	Typhoon	The resilience index is proposed using a probability of failure and fragility curve.	[227]
	Earthquake	Probability of damage state using fragility curve.	[228]
	Ice disasters	Introduced expected value of system resilience and component resilience index (pre/post event).	[229]
	Typhoon	A system-level resilience index is proposed using the expected value of the minimum load reduction and the probability of the occurrence.	[230]
	Short circuit faults	Proposed source path destination index, branch vulnerability index, loss of load index.	[231]
	Hurricane	Introduced resilience achievement worth using redundancy, robustness, and responsiveness.	[1]
	High wind	Measured Loss of Load Frequency (LOLF) and Loss of Load Expectation (LOLE).	[53]
	Severe weather	Proposed the FLEP resilience index, where the metrics are derived from the state, such as how fast, low, extensive, and promptly behave.	[19]
	Severe weather	Introduced electrical degree centrality, betweenness, closeness centrality, and network efficient centrality indices to respond to extreme events.	[232]
Distribution	Severe weather	proposed WRAP index, i.e., withstand, recover, adapt, and prevent index as a resilience measurement.	[18]
	Severe weather	Proposed recovery resilience index using total restoration and lost revenue impact factor.	[233]
	Severe weather	Introduced topological resiliency by considering electrical betweenness, algebraic connectivity, and clustering coefficient	[234]
	Unexpected disaster events	Proposed a resilience index in terms of expected energy curtailment during a disturbance considering equal probabilities of event times.	[34]
	Impact on fault	Proposed a system resilience index by considering total energy supplied to critical loads and average voltage variation.	[13]
	Severe weather	Introduced a resilience index considering customer outage duration, load not served, and restoration rates.	[235]
	Cyber-attack	Introduced a resilient distributed state estimator to withstand cyber-attacks in the distribution system.	[236]
	Cyber-attack	Proposed a cyber-resilient DC microgrid design subject to false data injection and time synchronization attacks.	[237]
	Severe weather	Proposed a resilience evaluation index with the consideration of load restoration strategy with multi-energy system coordination.	[238]
	Severe weather	Various resilience measurement indexes were introduced, such as expected probability of interruption, expected energy not served, and expected outage duration.	[239]
	Severe weather	Proposed vulnerability index and degradation index using a measure of performance curve.	[14]
	Severe weather	Introduced resilience index based on resourcefulness, rapid recovery, robustness, and adaptability.	[240]

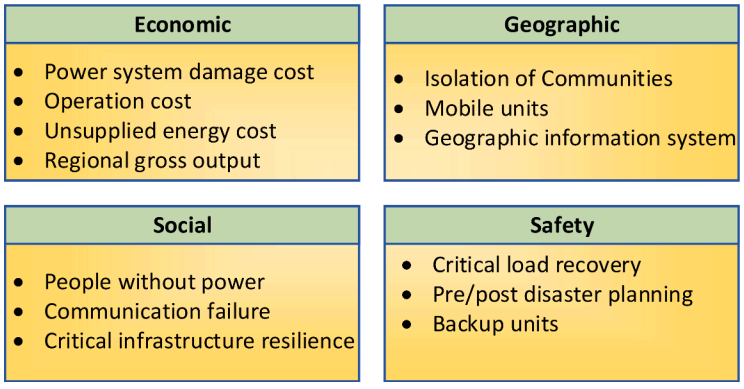


Fig. 16. Performance-based resilience concern.

generation sources from different zones to create a hybrid system that offers reliable and resilient power. At the transmission level, numerous resiliency approaches, such as decentralization, reconfiguration, hardening, and ancillary support, must be considered. Similarly, in the distribution system, smart devices, security features, demand response, and emergency services are equally important to handle extreme events [270]. The various methods at different levels of the power system are responsible for improving resilience and minimizing the diverse impact on the economy and society, as discussed in TABLE VII.

3.3. Methods of resilience analysis

There are various methods of analysis and quantification of resilience, such as scenario-based, fragility-model, probability risk assessment, vulnerability, failure mode, graph theory, adaptiveness/responsiveness/robustness, game theory, and economic impact. A comprehensive analysis of power system resilience can be done by selecting suitable combinations of these models concerning their operational and strategic aspects, which are discussed in TABLE VIII.

Table VII
Measures to improve resilience at different level.

Level	Classification	Methods to improve resilience
Generation	Traditional plant	Comprehensive assessment[271], Unit commitment [272], Coordinated generation [273], Multi-area power system [274], Energy supply chain[275]
	Distributed energy generation	Integrated assessment [239], Hybrid system integration [34], Adaptive control system [276], Efficient power converter [277] DER scheduling [278], Optimal allocation [197], Remote monitoring [279], Diesel generator scheduling [280], Vehicle-to-grid integration [281]
	Energy storage	Fixed/mobile storage scheduling [280,282], Large-scale energy storage [283] Enabling various energy storage solutions such as pumped hydro [284], compressed air energy [285], flywheel [286], superconducting magnetic energy [287], hydrogen energy [61], and more.
	Integrated approach	Proactive maintenance planning and scheduling [243], Automatic generation control [288], Interconnection with the energy market [289], Cybersecurity adoption [290], Contingency planning [291]
Transmission	Decentralization	Microgrid (preventive and proactive planning) [117,292], Interconnected microgrids [293], Integrate with the energy market and transmission system operator [294], Adaptive protection system [295], Modular substation [296], Remote monitoring of fault [297].
	Ancillary support	Flexible alternating current transmission systems (FACTs) [288], Protection devices [298].
	Reconfiguration	Tie-line placement [299], Islanded connection facility [300], State-based approaches [301], and Radiality constraint investigation [302].
	Hardening	Fragility modeling of poles and towers [1], Vegetation management [7], Undergrounding [303], Elevated substations, and choosing less prone to flood [304].
Distribution	Smart devices	Smart transformers [305], Smart meters [306], Smart reclosers and Smart sensors [307], Distribution automation [308], Intelligent electronic devices [309], Internet of Things (IoT) devices [310], Disturbance detection systems [311]
	Demand response	Energy scheduling [312], Load scheduling [313], EV scheduling [314] Restoration scheduling [315], Load shifting/shedding [316], and Demand bidding and clearing [317].
	Security	Cyber-resilient scheme [318], Blockchain technology [289].
	Emergency Preparedness	Mobile storage/diesel units [319], Mobile substation [320], Mobile EV charging infrastructure [202], Mobile Crew [321].

Table VIII
Methods of resilience analysis and quantification index.

Analysis	Models used	Measured indices	Ref.
Scenario-based	Monte Carlo Simulation (MCS)	Expected energy not served (EENS)	[322]
	Mixed Integer Linear Programming (MILP)	Post-disaster recovery	[323]
	Risk-based contingency analysis tool	Vulnerability and criticality	[324]
	Probabilistic hazard	Impact of hazard, service demand	[325]
Fragility-model	MCS	Average total power generated and average total power consumed	[326]
	MCS	Reliability Achievement Worth	[1]
	Stochastic robust optimization	Recovery, expected repair time	
	A detailed review	–	[327]
Probability risk assessment	MCS	Probability of failure	[1]
	SAPHIRE and PowDDER	System failure, probability of failure, and adaptive capacity	[328]
	Ground-motion prediction model	Criticality ranking, seismic risk assessment	[254]
	Rare events logistic regression	Failure Probability	[329]
Vulnerability	A detailed review	–	[330]
	MCS	CvaR, EENS, and interruption cost	[254]
	Rankine vortex model	Vulnerability assessment index	[331]
	Fragility	Failure probability and vulnerability rate	[332]
Failure mode	A systematic review	–	[333]
	Risk-based FMEA	failure modes, fault tree	[334]
	Stochastic state transition	Assortativity coefficient and largest amount of power supply	[335]
	MCS	Load loss and failure rate	[336]
Graph Theory	Coalitional game theory	Expected load curtailment and expected critical load recovery	[337]
	Graph theory	Weighted load shedding	[338]
	Hetero-functional graph theory	out-degree centrality and targeted disruption	[339]
	Graph theory and Choquet integral	Critical load recovery	[340]
Adaptiveness/ Responsiveness/ Robustness	Multi-stage hybrid-stochastic and robust model	Senstivity, adaptive investment	[341]
	Dynamic-adaptive load shedding	Voltage stability index and rate of change of frequency	[342]
	robust mixed-integer optimization	preventive response and Emergency Response	[29]
	Stochastic robust optimization	System resilience	[343]
	MILP	Load shedding, EENS, and planning cost	[344]
	MILP	Total scheduling cost	[64]
	Coordinate descent and line search	Contingency	[345]
Game Theory	Attacker-defender model	Load loss [346]	[346]
	Information gap decision theory	Length of outages and cost	[347]
	Stackelberg game-based	Sensitivity and cost	[348]
	Collaborative game theory	Social welfare	[349]
Economic	A detailed analysis	Economic models	[350]
	Contingent valuation survey	Value of lost load and resilience decision-making	[351]
	Meta-heuristic algorithm	Techno-socio economic	[352]

4. Power system resilience in global perspectives

With reference to the power system resilience study, the U.S. has developed key technologies and frameworks to combat extreme events. For instance, the U.S. Department of Energy (DOE) has created a resilient framework for a multi-hazard approach. Similarly, several institutions have made significant progress in resilience research, including Sandia National Lab, Pacific Northwest National Lab, Oak

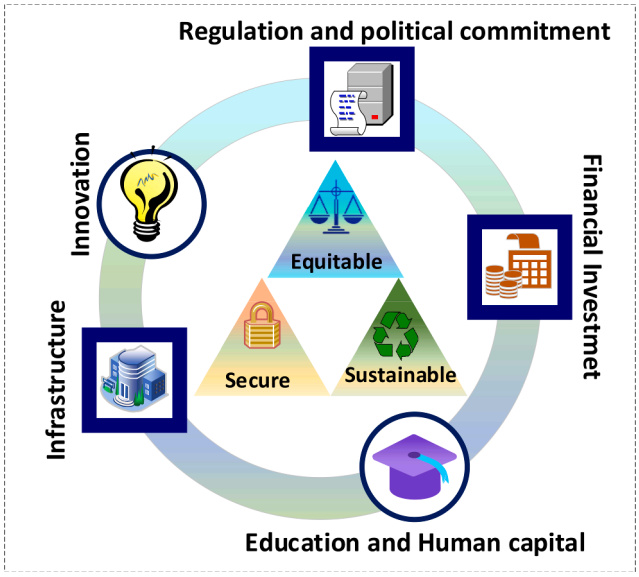


Fig. 17. Energy transition index-framework [357].

Table IX
Greenfield and brownfield comparison.

Approach	Pros	Cons
Greenfield	Innovation	Higher initial cost
	Customization	Uncertainties
	Optimized efficiency	Risk to return
	Environmental consideration	Project duration
	Highly scalable	
	Future proof	
Brownfield	Cost savings in initial design	Integration challenges
	Adaptability	Scalability
	Quick start	Limitations
	Existing resources used	Environmental impact
	Mitigated business risk	Potential disruptions

Ridge National Lab, Argonne National Lab, and others [353]. While the U.S. has contributed substantially to resilience study, other countries have also contributed significantly to building a resilience framework. In Australia, bushfires are a common type of extreme event, and to respond to that, the Bushfire Cooperative Research center (CRC) was established in 2013, which proposed many restoration strategies to boost resilience. In addition, the Australian Energy Market Operator (AEMO) and Commonwealth Scientific and Industrial Research organisation (CSIRO) have developed many disaster mitigation frameworks against heat-waves, cyclones, and cyberattacks. In the U.K., Energy System Catapult, center of Energy Informatics, and National Grid Plc have advanced resilience knowledge and proposed many restoration plans. In Europe, the European Commission has funded billions of Euros to progress the study on resilience on account of extreme events. There are other institutions in Europe, like the European Network of Transmission System Operators for Electricity (ENTSO-E), Electric Power Research Institute (EPRI) Europe, European Distributed Energy Resources Laboratories (DERlab), center for Research and Technology Hellas (CERTH), and National Renewable Energy center (CENER) are also participating in resiliency study. In China, there are several institutions and labs working towards building a sustainable and resilient power system. Some of these are State Grid Energy Research Institute (SGERI), China Electric Power Research Institute (CEPRI), State Grid Smart Grid Research Institute (SGSRI), China National Renewable Energy center (CNREC), China Southern Power Grid Research Institute (CSGRI) and state key laboratories. Similarly, in India, Central Power Research Institute (CPRI), National Institute of Wind Energy (NIWE), Power System Operation Corporation Limited (POSOCO), Bureau of Energy Efficiency (BEE), and a few more institutions are greatly involved in resilience and research and focusing on energy transition and NetZero target.

This progress encompasses grid modernization with microgrid and smart grid technologies, cyber-resilient technology, black start capability using energy storage units, and emergency response procedures (for example, utilizing mobility units). However, it's essential to note that this research and development primarily occur within the national context. Energy systems worldwide confront significant challenges influenced not only by natural disaster contexts but also by societal, economic, geographic, and other environmental reflections [354]. Considering resilience in a broader global context, collaboration and experience sharing become vital [355]. With this broad and collaborative approach, a more robust and adaptable resilient framework can be designed to address extreme event challenges not only in a particular geographic location but also globally [356]. So, a global resilience partnership could significantly benefit the world by reducing the power system damage cost, minimizing life threats, and helping to boost the economy. In addition, it can help to achieve the NetZero target and sustainable future.

Fostering the future efficient global energy transition, a few forefront institutions are involved in the discussion of energy sustainability, security, and equity in international forums such as the World Energy Council and the World Economic Forum. Notably, to strengthen the energy sector towards reliability, security, and resilience, some major factors must be considered. These factors, including innovation, education, human capital, infrastructure, regulation and political commitment, and financial investment, can be seen in Fig. 17. The detailed elaboration of each factor is discussed in [358]. Since 2010, the World Energy Council has introduced a metric called the energy trilemma index, which can measure the country's energy performance by considering energy sustainability, security, and equity [357]. With this metric, the nation can highlight the challenges and opportunities in balancing the index to meet the present and future energy goals. In fact,

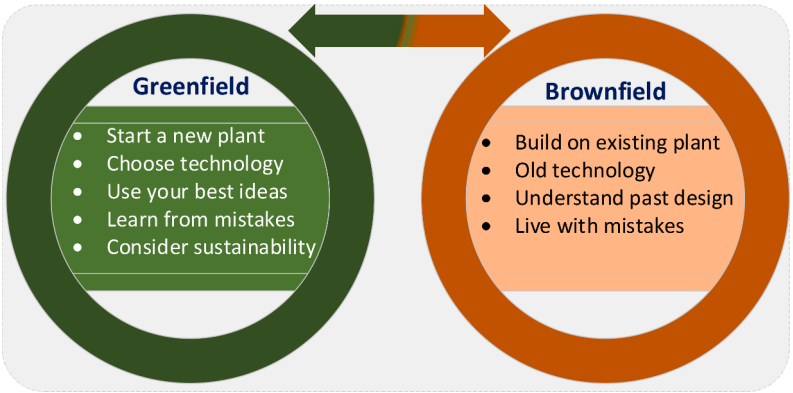


Fig. 18. Greenfield vs Brownfield.

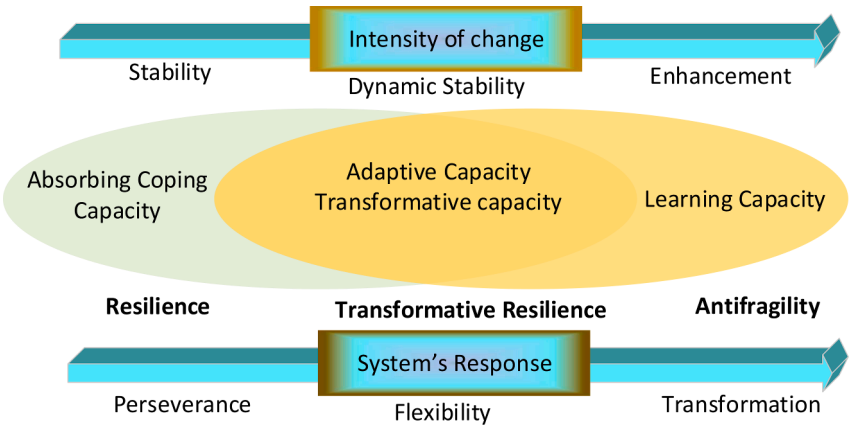


Fig. 19. Resilience, transformative resilience, and antifragility [370].

the energy trilemma index could help show the availability of energy resources in a particular nation.

On the other hand, enhancing the energy sector's resilience involves a multifaceted approach that requires participation from various stakeholders and consideration of the political climate [359]. For instance, stakeholder engagement is crucial, as there are numerous utility companies playing a vital role in developing and maintaining the infrastructure. In addition, technology providers could significantly contribute to improving the system's resilience by incorporating advanced tools to handle extreme events [360]. Furthermore, the political climate regarding the policy framework is equally important [361]. These include long-term planning with specific objectives, seeking international cooperation, prioritizing investments with incentives, educational programs, and research development. This engagement ensures a comprehensive plan for resilience enhancement, considering technical, regulatory, and social aspects.

5. Shaping resilient futures for power systems

The power system infrastructures are undergoing a significant transformation through greenfield and brownfield projects [362]. These two approaches are developed with distinct considerations in terms of design and approaches. Considering the Greenfield approach, they start with a Greenfield, creating new infrastructure. For example, setting up a new power plant with the adoption of advanced technologies, considering reliability, sustainability, and resilience [363]. In contrast, the Brownfield approach is an extension of the existing power system, also known as power system restructuring, that can be modernized to adapt to extreme event situations. Distinguishing the implications of both approaches in the power system context is vital, as the energy sector needs resilient solutions to minimize economic loss and life threats.

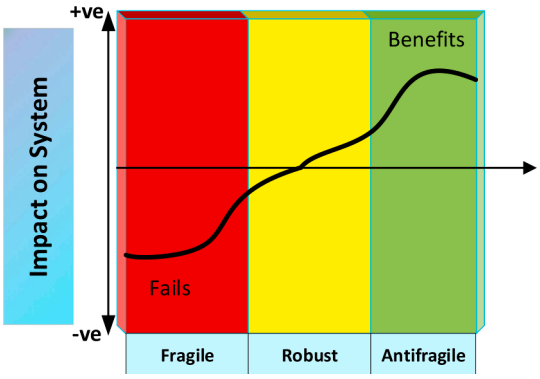


Fig. 20. Antifragility curve.

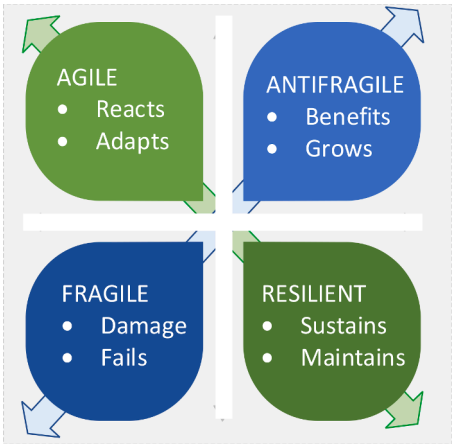


Fig. 21. Extending resilience to antifragility.

Indeed, in brownfield approaches, some technical challenges might be encountered. For example, existing technologies could face integration challenges due to advanced technology that might not fit or adapt to the existing technology. These challenges include compatibility problems, interoperability issues, system legacy concerns, data security issues, scalability challenges, cost implications, and more [364]. So, it can be a minor breakthrough. On the other hand, greenfield approaches can be significant and designed according to stakeholders' requirements, geographical uncertainties, and weather conditions. However, the hybrid approach envisions the critical combination of new establishments with existing power systems and turns into a keystone for a sustainable and resilient energy future.

As we move towards green and clean energy, a greater number of decentralized plants have been established in the last few decades, which use renewable energy resources to form a mini-grid, microgrid, and nano-grid. These approaches bring numerous benefits to the stakeholders, end-users, and societies. For example, in the wake of extreme events, the centralized plant may be affected leading to power outages for millions of customers. However, decentralized plants can only affect nearby areas; other resources can be isolated and supply other areas. Furthermore, redeveloping existing plants using hybrid design approaches can also be fruitful, fostering sustainable urban development and smart growth [365]. The pros and cons of Greenfield and Brownfield can be seen in TABLE IX, and the viewpoint can be seen in Fig. 18.

Furthermore, adopting new technologies in the power system is key to improving resilience. However, it has a broad range of advantages and disadvantages on account of economic factors [24]. Resilient

design-based power systems can save trillions of dollars annually as they can minimize the outage cost and reduce damage to the electricity infrastructure [366]. However, adopting new technologies can increase the huge investment cost, which can financially burden utility companies and end-users in the short term. In the long run, it could be beneficial as a resilient system provides continuous power supply during normal and disruptive event conditions [367].

6. Innovation of resilience and its beyond: antifragility

Many state-of-the-art technologies have been developed in the past few years to enhance power system resilience against natural disasters, cyber-attacks, and large faults due to component failure. A few of these breakthrough technologies include microgrids and smart concepts, distributed generation, large-scale energy storage, power system interconnection using renewables, smart devices, solid-state transformers, islanding and black start capabilities, cybersecurity protocols, wide-area monitoring and control, demand-side management, EV integration, FACTS devices, as can be seen in TABLE VII. Additionally, numerous studies have proposed resilience indices for different levels of the power system, considering diverse sources of extreme events. However, no standard resilience index has been developed yet, and it is still in the research and development stage. New emerging technologies are needed to advance the resilience enhancement approach and are currently under development. For example, intelligent devices or sensors could respond to the system in terms of fault detection and mitigation, artificial intelligence-based anomaly detection and prevention, probabilistic assessment, multi-layer resilience schemes, and more. Considering the advancement of energy storage, hydrogen and solid-state battery storage are emerging technologies that will soon greatly impact resilience and sustainability. Similarly, other exciting R&D technologies are also fast approaching, such as quantum computing for grid management, Blockchain technology for securing the energy market, digital-twin technology, etc. While the energy sector is currently focusing on innovative technologies to improve resilience, a few researchers are thinking beyond the concept of resilience, called antifragility, which is discussed in the next subsection.

In the last few years, energy sectors have undergone a significant transformation to enhance the system's resilience to cope with extreme events. However, an innovative concept has emerged from a transformative idea extending the conventional resilience philosophy called antifragility [368]. Nassim Nicholas Taleb introduced this concept in his



Fig. 22. Challenges and opportunities for power system resilience.

book, named "Antifragile: Things That Gain from Disorder" [369]. Indeed, resilience focuses on the capacity to tolerate disturbances and recover from them. Antifragility goes a step further by suggesting that some systems may not only resist disturbances and uncertainties but also thrive in the midst of disasters. Fig. 19 demonstrates the transition from resilience to antifragility in terms of stability and flexibility.

An antifragile system primarily improves positive outcomes while considering uncertainties in a random environment that goes beyond initial estimations. Recently, Coppitters et al. applied the antifragility concept in energy system design, considering different uncertainties of renewables and identifying the antifragile metric as the design alternative [371]. A few studies have been carried out on fragility modeling in the energy system; it needs further exploration in this review paper, as discussed below.

The fundamental dimensions of resilience comprehend stability, adaptability, and recoverability. The concept of antifragility has evolved to extend beyond traditional resilience dimensions. This extension, known as transformative resilience, implies embracing both adaptive and transformative capacities. Furthermore, learning capacity can be distinguished by the growth and innovation derived from resilience concepts. This introduces a novel perspective on sustainability and productivity within non-linear and multidimensional adaptive systems [372].

Furthermore, the impact on the power system can be measured by three factors: fragile, robust, and antifragile [373]. Considering Fig. 20, if the system is in the robust zone, the performance of the system aligns with the design expectations. If the curve shifts toward the left from the robust zone, referred to as the fragile zone, it forces the performance to decline, leading to a system failure and an inability to respond to events. Shifting toward the right from the robust zone, known as the antifragile zone, results in positive outcomes similar to growth and innovation. Moreover, the extension of resilience from agile to resilient and from fragile to antifragile concepts is illustrated in Fig. 21. An antifragile model can also benefit in various ways, including dynamic adjustments, performance improvements [374], adaptability to renewable uncertainties [371], reduced downtime [375], risk mitigation, and more cost-effective systems [370].

7. Challenges and opportunities

The electric infrastructure is essential to modern society, and its resilience is vital to our economic and social well-being. However, it is also vulnerable to various challenges threatening its resiliency. As the electric infrastructure continues to evolve, plenty of opportunities can make the power system as resilient as possible. Some specific challenges and opportunities are highlighted in Fig. 22.

8. Conclusion

Implementing a resilient structure in the power system network and the enhancement method has become a key necessity. In response to natural disasters and man-made attacks, resilience is acknowledged as one of the emerging research trends essential for critical infrastructure and the whole community. This paper has provided a detailed review of the current state of knowledge in power system resilience. The paper initially explored the fundamental concept of resiliency with the performance curve followed by four resilience enhancement pillars: hardening, distributing, smartening, and building. Afterward, the review study extends to the resilience quantification indices and their measures. Further, the discussion extends to resilience from global perspectives, and then innovation of resilience for futuristic solutions is explored. Finally, the challenges and opportunities of power system resilience are discussed, posing insights into trends and limitations. By grasping the challenges and opportunities, investors in the energy sector can work towards building a resilient electric infrastructure that can cope with extreme events, adapt to varying conditions, and ensure a reliable and sustainable power supply to meet the needs of society.

The following recommendations for future work can be made in light of this review study on power system resilience:

- A comparison study on the resiliency of systems other than electric grids is a very interesting idea to extend the research to benefit the power system community with potential resiliency solutions in other complex systems.
- Deeper analysis of the challenges and opportunities, including political climate and global framework.
- Advanced resilience quantification approaches.
- Cross-domain collaboration includes cyber-security experts and disaster response agencies.
- Scenario planning, probability and risk assessments, and antifragility modeling on account of extreme events.

CRediT authorship contribution statement

Dillip Kumar Mishra: Conceptualization, Writing – original draft. **Mohsen Eskandari:** Formal analysis, Writing – review & editing. **Mohammad Hossein Abbasi:** Writing – review & editing. **P. Sanjeevikumar:** Supervision, Writing – review & editing. **Jiangfeng Zhang:** Supervision, Writing – review & editing. **Li Li:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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